

MONCHO UNSCHOOLING

Basketball

Unit Study

“What if the whole world, from the physics of a bounce to the story of a peach basket, were hiding inside one orange ball and a hoop?”

Physics | Math | Movement & Body | History | Geography | Language Arts | Life Skills & SEL | Art & Creativity | Music | Biology

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

🔬 PHYSICS

One orange ball and a hoop are a whole physics lab: bouncing, flying, and falling all happen in a single shot.

🏀 Challenge 1: The Bounce Lab

🎯 Measuring energy

If you drop a ball from twice as high, will it bounce back twice as high? ❓

You're the ball-lab tester, and today you find out how much bounce a basketball keeps.

1. Tape a tape measure or yardstick to a wall, starting at the floor.
2. Hold the ball so its bottom is at 20 cm (or 8 in). Drop it, and have a friend watch the bottom of the ball at the top of its first bounce. Film it in slow motion if you can.
3. Repeat 3 times and write down your best guess each time.
4. Do the same from 60 cm (2 ft) and 100 cm (3 ft).
5. On a sheet of paper, make a table of drop height and bounce height. Then write your own conclusion: does the ball keep the same fraction every time?

Why: some of the ball's energy turns into sound and a little heat each time it squishes against the floor, so it can never bounce back to where it started.

🌱 **Make it Easier:** Use just two heights and compare them by eye: is the bounce a little less than the drop, or a lot less?

🚀 **Level it Up:** For each drop, divide the bounce height by the drop height and compare your answers. Then try a softer, less-pumped ball and see how the numbers change.

🎯 Challenge 2: Arc Showdown

🎯 Projectile motion

Is it easier to sink a shot that flies flat and fast, or one that floats up high?❓

Set up a hoop, or a bucket or laundry basket on the floor, about 2-3 meters away. You'll shoot 10 times with each of three arcs:

1. Flat arc: throw it low and hard, almost straight at the target.
2. Medium arc: send it up about as high as your head.
3. High arc: lob it way up, higher than the top of a door, so it drops almost straight down.

Draw three columns on paper (flat, medium, high). After every shot, add a tally mark under "made" or "missed" for that arc. Take your shots from the same spot each time so the test is fair.

Why: a ball coming down steeply sees a bigger opening, because from above a hoop looks like a wide circle, but from the side it looks like a thin slit.

🌱 **Make it Easier:** Take only 5 shots per arc and stand closer to the target.

🚀 **Level it Up:** Move back one big step and repeat all three arcs. Do the best arcs stay the best when you're farther away?

🥄 Challenge 3: Build a Mini Shooting Machine

🎯 Levers and launching

How can a rubber band and a spoon throw something farther than your arm can? ❓

You'll need 6-8 craft sticks, 2-3 rubber bands, a plastic spoon, tape, a pom-pom, and a small paper cup for your hoop.

1. Stack 5 craft sticks and wrap a rubber band tightly around each end to make a thick base.
2. Stack 2 more sticks and wrap a rubber band around one end only. Slide the stack open to make a V, with the free ends apart.
3. Tape the spoon to the top stick of the V so the bowl hangs past the edge.
4. Place the pom-pom in the spoon. Press the spoon down, then let go to launch it.
5. Set your cup 1 meter away. Move the cup or change how far you press the spoon until the pom-pom lands in it.

Why: pressing the spoon stores energy in the stretched rubber band, and letting go turns that stored energy into motion.

🌱 **Make it Easier:** Skip the V and just tape the spoon to a stack of sticks, then press and release. Put the cup right in front of it.

🚀 **Level it Up:** Find out how many centimeters the spoon must press down to land in the cup at 1 m, then predict how far to press for 1.5 m.

📐 MATH

Every number in basketball, from a 10-foot hoop to a 94-foot court, is waiting for you to measure it, multiply it and use it to win.

📏 Challenge 1: Tape the Hoop!

🎯 Measuring and comparing

How many of you would have to stand on each other's heads to reach a real basketball hoop? ❓

A regulation hoop sits 10 feet (about 3.05 m) above the floor. Let's find out how you measure up.

1. Stick a strip of tape on a wall and measure up from the floor with a tape measure, marking 10 feet (305 cm) with a long piece of tape.
2. Stand with your back against the wall and mark your height with a small piece of tape. Write it in feet and in centimeters.
3. Now stretch one arm straight up and mark your full reach. Write that in feet and centimeters too.
4. Subtract: how many feet and how many centimeters short of the hoop are you? Then measure the rest of your family and do the same.

Why: Your reach is taller than your height because your arm adds extra length above your head, which is why players always jump from their reach, not their height.

🌱 **Make it Easier:** Measure only in centimeters and just find who is closest to the tape.

🚀 **Level it Up:** Figure out how many of you stacked head to toe, using your heights added together, would be needed to reach the hoop, then check if your family's total is enough.

🧮 Challenge 2: The Court Surveyor

🎯 Area and perimeter

What happens to the size of a court when you cut both its length and width in half? ❓

You're the court surveyor, hired to draw a smaller court for the neighborhood. An NBA court is 94 feet long and 50 feet wide (about 28.7 m by 15.2 m). Your job is to cut it down to half the length and half the width.

1. Calculate the real court's area (length × width) and perimeter (add all four sides).
2. Halve each side: what are the new length and width?
3. Calculate the new area and perimeter.
4. Using a tape measure, sidewalk chalk and a driveway or sidewalk, mark out your half-size court. Count your steps to help: one step is about 2 feet.
5. Compare: is the new area half of the old one? Is the new perimeter half?

Why: Perimeter is a length, so it halves, but area multiplies two lengths together, so it shrinks to one quarter.

🌱 **Make it Easier:** Use a smaller court of 20 feet by 10 feet and only find the perimeter by adding the sides.

🚀 **Level it Up:** Split your chalk court into a half court with a line and calculate the area of just one half, then work out how many players could stand in it with 4 square feet each.

🏀 Challenge 3: Shoot for Points

🎯 Probability and strategy

Is it smarter to take the easy shot or the hard one that's worth more? ❓

In the NBA, the three-point line is 23 feet 9 inches (about 7.24 m) from the basket at the top, and 22 feet (about 6.7 m) in the corners. Set up your own version and see which shot pays off.

1. Use chalk or tape to make two lines: a close line worth 2 points and a far line worth 3 points. Use a bucket or box as the basket.
2. Each player takes 10 shots, 5 from each line, and a scorekeeper records every make and miss.
3. Calculate each player's points from 2-point shots and from 3-point shots.
4. Work out the percentage you made from each line: makes ÷ shots × 100.
5. Compare: if you only shot from one line next round, which one would give you more points in 10 shots? Write your choice and play a second round to test it.

🌱 **Make it Easier:** Count only makes and compare which line had more, without calculating percentages.

🚀 **Level it Up:** Find the break-even point: if you make 6 of 10 from the 2-point line, how many of 10 do you need from the 3-point line to score the same points?

🏃 MOVEMENT & BODY

Basketball turns your whole body into the game: hands, feet, balance, and brain all working together with one orange ball.

🏀 Challenge 1: The Shoe Slalom Dribble

🎯 Ball control

Which of your hands is secretly the boss of the ball, and can the other one catch up?❓

Time to build a dribbling course and chase your own personal best.

1. Line up 5 shoes in a row, about one big step apart.
2. Stand still and dribble 20 times with your right hand, then 20 with your left. Which hand feels wobblier?
3. Now weave through the shoes with a figure-eight dribble: bounce the ball with one hand, step around a shoe, switch hands, and keep going.
4. Remember the rule: once you catch the ball with both hands and stop bouncing it, you can't start bouncing it again. That's a "double dribble" violation, so keep it bouncing the whole way!
5. Time yourself with a clock, run the course 3 times, and write down your best time on a scrap of paper.
6. Try to beat your own record, then do it again with your weaker hand.

🌱 **Make it Easier:** Use only 3 shoes, spread them wider apart, and use a soft playground ball. Forget the timer and just count how many shoes you pass without losing the ball.

🚀 **Level it Up:** Add a shoe or two, bring them closer together, and dribble the whole course without looking down at the ball. Add a challenge: dribble the course backward.

💃 Challenge 2: Ball-Free Footwork Dance

🎯 Footwork and balance

Can you play basketball moves with no ball at all and still look like a pro?❓

Basketball players spend as much time moving their feet as moving the ball, so let's turn their footwork into a dance.

1. Learn 3 real moves: a pivot (one foot stays glued to the floor while the other foot swings around), a jump stop (jump and land on both feet at the same time), and a defensive slide (stay low and slide sideways without crossing your feet).
2. Invent 2 more moves of your own, like a spin, a fake, or a silly victory shuffle.
3. Put your 5 moves in an order you like and practice them 3 times until they flow.
4. Pick a favorite song, give your dance a name, and perform it for your family.

Why: Pivoting on one foot lets a player turn and look for a pass without traveling, and staying low in a defensive slide keeps your balance close to the ground so you can change direction fast.

🌱 **Make it Easier:** Dance with just 3 moves, and count out loud "1, 2, 3" for each one. Practice slowly before adding the music.

🚀 **Level it Up:** Add a 6th and 7th move, then do the whole routine to a faster beat. Try teaching it to a family member so you can dance it together.

♿ Challenge 3: Seated Shooting Challenge

🎯 Balance and coordination

What would it feel like to play basketball when your legs can't help you?❓

Wheelchair basketball began in the 1940s, played by veterans, and became a Paralympic sport at the first Paralympic Games in Rome in 1960. Let's feel what those players do every game.

1. Set a bucket or laundry basket on the floor or a low table about 2 meters away. Use a soft ball or a pair of rolled-up socks.
2. Sit on a sturdy chair with your feet lifted off the floor, so your legs can't push you up. Take 10 shots at the basket and count how many go in.
3. Now stand up and take 10 more. Does it feel different?
4. Now try moving like a wheelchair player: sit on a smooth floor on a towel, and use your hands to push yourself and turn. Notice how much your arms and core must do.
5. Write or draw a quick note about what was hardest and what surprised you.

Why: Without legs to push upward, a seated shooter gets all the power from the arms, shoulders, and core, which is why players train those muscles so much.

🌱 **Make it Easier:** Sit closer to the basket and use a bigger target, like a laundry hamper. Take only 5 shots.

🚀 **Level it Up:** Add a passing partner: sit across from a friend or family member and pass the ball back and forth 10 times, then shoot. See how your teamwork changes when everyone is seated.

📜 HISTORY

Every great sport started with someone's idea, and basketball started with a gym teacher, a ball, and a couple of peach baskets.

🧺 Challenge 1: The First Game, Backyard Edition

🎯 Historical Recreation

What would it feel like to play a game on the very first day it existed? ❓

You're James Naismith, the gym teacher. It's December 1891 at a YMCA training school in Springfield, Massachusetts, and your students are bored. You need a new indoor game, and you have a ball and a hoop idea.

1. Tape or tie a laundry basket high up on a wall, fence, or doorframe (with an adult's help). Naismith's baskets hung on a gym balcony.
2. Find a soccer ball or any ball you can throw.
3. Make a few of your own simple rules: no running while holding the ball, and everyone can pass in any direction.
4. Invite a friend or family member and play a short game to 3 baskets.

🌱 **Make it Easier:** Use a bucket on the floor instead of hanging it up high, and take turns tossing from a line.

🚀 **Level it Up:** Add one more rule, play again, then decide whether your new rule helped or made things worse.

⚖️ Challenge 2: Rule Detective

🎯 Comparing Rules

If you could change one rule from 1891, which would the game need most? ❓

Naismith wrote 13 original rules for his new game. Some are still used today, and some are very different!

1. Look up "Naismith's 13 rules of basketball" in a book or with a grown-up's help online.
2. Pick 3 rules that surprise you, such as the ball being moved only by passing, with no dribbling.
3. For each one, find out how the rule works in basketball today.
4. Hold a family debate. Each person votes for the one rule change they think made the game better, and gives one reason.

Why: Games change when players discover problems, like a ball bouncing out of reach or players crowding the basket, and the rules are updated to fix them.

🌱 **Make it Easier:** Compare just 2 rules and vote with a thumbs-up or thumbs-down.

🚀 **Level it Up:** Invent your own "rule 14" for a game you play, then test it with your family.

🏆 Challenge 3: Women Who Changed the Game

🎯 Research and Presenting

How did women's basketball go from a small beginning to playing in front of huge crowds? ❓

The WNBA (Women's National Basketball Association) began playing in 1997. Before that, women's basketball had to grow step by step, including a big moment at the 1976 Olympics.

1. With a grown-up, look in a library book or website for what happened in women's basketball at the 1976 Olympics.
2. Find one player or team from that story and write down three facts.
3. Look up when the WNBA started and name two things that changed after it began.
4. Ask a family member what they know about women's basketball, and write down what they say.
5. Stand up and share a 1-minute mini-report with your family member.

🌱 **Make it Easier:** Research just one fact about the 1976 Olympics and one about the WNBA, then say them aloud.

🚀 **Level it Up:** Ask your family member 3 questions about women's sports when they were young, then compare their answers with what you found.

🌍 GEOGRAPHY

Basketball has traveled from one gym in Massachusetts to courts on every continent, and a map can show you how far it has gone.

🗺️ Challenge 1: Pin the Players

🎯 Reading world maps

If you stuck a pin in every country where a famous player was born, what pattern would the map reveal? ❓

Grab a world map (paper, or print one), a pencil, and some stickers or thumbtacks.

1. Find Shanghai, China, and pin it. Yao Ming grew up there and later played for the Houston Rockets in the NBA. He stood about 7 feet 6 inches (about 229 cm) tall.
2. Look up 5 more famous players and pin the country each one comes from.
3. Put a different sticker in New York City for Rucker Park, a famous outdoor court.
4. Use a pencil or string to trace the path from Shanghai to New York. Which oceans does it cross?
5. Look at your pins. Which continents have the most? Which have none?

🌱 **Make it Easier:** Pin just 3 players, and use a different color for each continent so it's easy to count.

🚀 **Level it Up:** Guess which continent will have the next big wave of players. Then research one country that has few pins and find out whether basketball is growing there.

🏙️ Challenge 2: The Basketball Highway

🎯 Distance and landforms

What do five far-apart cities that love basketball have in common on the map? ❓

Pick 5 cities that love basketball: Chicago, Manila, Belgrade, Lagos, and Buenos Aires.

1. Open Google Earth or a paper atlas and drop a pin on each city.
2. Lay a piece of string between two cities, then hold it against the map scale to measure the distance. Do this for at least 5 pairs.
3. Find the nearest ocean, river, or mountain to each city.
4. Sketch a tiny postcard of that landscape for each city.
5. Predict which city has the most outdoor courts for its size and why land and weather might matter. Then look up a photo of a real court in each place to check.

Why: A map scale is a promise that a short length on paper equals a fixed real distance, so string pulled straight can be turned into kilometers.

🌱 **Make it Easier:** Use only 3 cities, and measure just the distance from your home to each one.

🚀 **Level it Up:** Rank all the distances from longest to shortest. Then find which of the five cities are in the Northern Hemisphere and which are in the Southern Hemisphere, on opposite sides of the Equator.

☀️ Challenge 3: Court for Every Climate

🎯 Climate and design

Would the perfect outdoor court look the same in a rainy city, a desert, and a snowy town? ❓

Streetball and pickup games are played outdoors all over the world, including at Rucker Park in Harlem, New York City. Your job is to design courts for three very different places.

1. Pick 3 countries with different climates, such as one hot and dry, one rainy, and one cold and snowy.
2. For each, find the typical weather: summer and winter temperatures, and how much rain or snow falls.
3. Draw a court for each place on its own sheet of paper.
4. Label at least 3 design choices on each drawing, such as shade, drainage, floor material, or a roof.
5. Explain in one sentence why each choice fits that climate.

Why: Water pools on flat ground, so courts built where it rains a lot are usually tilted slightly so water runs off instead of making a slippery puddle.

🌱 **Make it Easier:** Design just 2 courts, one hot and one rainy, with 2 labels each.

🚀 **Level it Up:** Design a single court that works in all three climates, with parts that can change, like a roll-out shade or a drain you can open.

📖 LANGUAGE ARTS

Basketball comes with its own secret language, and every word in it has a story hiding inside.

📖 Challenge 1: Word Detective Dictionary

🎯 Word origins

Why do we call a basket "a swish" and a jump to the rim "a dunk"? ❓

Start your own basketball dictionary, with a page for each word.

1. Pick 5 words: swish, dunk, layup, dribble, rebound. Add others you hear while watching a game.
2. Write each word on its own page and draw a tiny picture for it.
3. Say the word out loud and listen. Does it sound like what it describes? Write down your guess for why it got its name.
4. Check a kids' dictionary, a library book, or ask a grown-up to look up the real origin. Copy it under your guess.
5. Staple the pages into a little book and add a cover.

Why: Some words are onomatopoeia, which means they copy a real sound. "Swish" sounds like the net moving when the ball passes cleanly through.

🌱 **Make it Easier:** Do only 3 words and just draw what each word looks like.

🚀 **Level it Up:** Find a word from another sport that comes from the same root and compare their stories.

🎙️ Challenge 2: Courtside Voice

🎯 Descriptive language

How can your voice make a simple shot feel like the most dramatic moment ever? ❓

You're the radio announcer, and the listeners at home can't see a thing. Only your words paint the picture!

1. Collect 5 big, juicy words first: "soaring," "unstoppable," "spectacular," "thunderous," "breathtaking."
2. Set up a shooting game at a hoop, a laundry basket, or a bucket. Every family member takes turns shooting.
3. Call each shot out loud, live, like a radio show. Mention the player's name, the ball's flight, and the crowd's reaction.
4. When someone shoots from far away, announce it like a legend: "From way downtown, just like Stephen Curry, one of the greatest three-point shooters in NBA history!"
5. Record your broadcast on a phone, or perform it live for an audience.

🌱 **Make it Easier:** Use only 3 big words and say one sentence per shot.

🚀 **Level it Up:** Add a second announcer who gives the "color commentary," explaining why each shot worked or missed.

📖 Challenge 3: Diary of a Brand-New Ball

🎯 Point of view

What would the very first basketball say about the day it was born? ❓

It's December 1891. You are the first basketball ever used in a game, and you have a lot to say!

1. Fold a paper in half and write "Dear Diary" at the top, or begin "Once upon a time, I was a ball."
2. Describe how it feels to be new and bouncy. Use your senses: what do you see, hear, and feel?
3. Add this surprise: the basket was closed at the bottom, so after every score someone had to climb a ladder to take you out. How would that feel, going in and getting stuck?
4. Show the ball's feelings with strong verbs (tumbled, stuck, squeezed, rescued).
5. Finish with how you feel about what comes next. Draw a small picture beside your words.

🌱 **Make it Easier:** Write just 4 sentences that start with "I am a ball and..."

🚀 **Level it Up:** Write three diary entries: the first day, a year later, and today, as the ball sees how the game has changed.

🤝 LIFE SKILLS & SEL

A basketball team only works when five different people learn to share the ball, keep their cool and keep getting better.

🏀 Challenge 1: Everybody Touches the Ball!

🎯 Teamwork

How does it feel when every single teammate gets a turn to matter? ❓

A basketball team puts 5 players on the court at once. Today you play your own mini-game with the same number of players.

1. Gather 4 friends or family members and a ball (a soft ball or rolled-up socks work too). Grab a hoop, laundry basket or box as your target.
2. Make one rule: all 5 players must touch the ball before anyone can shoot.
3. Set a timer for 4 minutes and play. Count how many baskets your team scores.
4. Play again for 4 more minutes. Did your passing get smoother?
5. Sit in a circle and talk: Who felt left out at first? What did a teammate do that made you feel included? What would you change next time?

🌱 **Make it Easier:** Play with only 3 players, and all 3 must touch the ball before a shot.

🚀 **Level it Up:** Add a second rule: no player can pass back to the person who just passed to them. Afterward, talk about how having extra rules changed your teamwork.

🎭 Challenge 2: Captain's Cool-Down

🎯 Managing Emotions

What would a great team captain say when everything goes wrong, and when everything goes right? ❓

You're the team captain, and a captain's words can lift a team up or drag it down. Act out three moments with a friend or family member, taking turns being the captain:

1. The missed shot: your teammate misses the winning shot and hangs their head.
2. The unfair call: the referee makes a call you really disagree with.
3. The big win: your team wins, and the other team is sad.

For each scene, first act it out the wrong way (grumpy, blaming, bragging). Then replay it the captain way: calm and kind. Notice how your voice, face and body change.

Now write a list of 5 calm-down strategies you could use in real life, like a slow breath in for 4 counts and out for 4, or a self-talk phrase such as "Next play." Circle your favorite one.

Why: when you're upset, your body speeds up and slow breaths tell your brain it's safe, which helps you think clearly again.

🌱 **Make it Easier:** Act out just one scene and pick only 2 calm-down strategies.

🚀 **Level it Up:** Add a fourth scene (a teammate who never passes) and write two sentences you could say to them that are honest but kind.

⏱️ Challenge 3: Beat the 24-Second Clock

🎯 Time Management

If you only had a few seconds to make a choice, how would you decide what to do first? ❓

In basketball, a shot clock limits how long a team can hold the ball before shooting. The NBA's is 24 seconds. You'll use that same pressure to design your own practice plan.

1. Pick one skill you want to improve (shooting, reading, drawing, a musical instrument, jump rope, anything).
2. Draw a week calendar with 7 boxes. In each box, write a small practice goal that takes 10 minutes or less.
3. Pick the 3 days you'll actually practice and decorate them as your "must-do" days.
4. Add a timed challenge: choose one 24-second task for your skill (for example, how many free throws, spelling words or jump-rope jumps can you do in 24 seconds?). Set a timer, try it, and write your score.
5. At the end of the week, try the 24-second challenge again and compare your scores.

🌱 **Make it Easier:** Plan just 3 practice days and a 10-second challenge instead of 24.

🚀 **Level it Up:** Add a "Backup Plan" box for what you'll do on a day when you feel tired or busy, so your habit doesn't stop.

🎨 ART & CREATIVITY

Every great basketball team starts with a look, a logo, a mascot, a show, and your imagination can invent all of it.

🏀 Challenge 1: Build Your Dream Team's Look

🎯 Logo Design

What makes a logo so powerful that you can recognize a team from across a stadium? ❓

Your imaginary team needs an identity before its first game. Grab paper, a pencil, markers, and a ruler.

1. Pick a team name and a mascot (a Thunder Llama? a Sneaky Fox?).
2. Sketch 3 tiny logo ideas, each in a box about the size of a postage stamp.
3. Circle your favorite and redraw it big, using only 2 or 3 bold colors.
4. Draw a simple jersey shape (like a tank top with a round neck) on a new sheet of paper.
5. Add your logo, a number, and your team name, then color it in.

Why: simple shapes and strong colors are easy for the eye to remember, even from far away or when a logo is shrunk small.

🌱 **Make it Easier:** Skip the three sketches. Draw one mascot, then color a jersey with just one color and your team name.

🚀 **Level it Up:** Design a second jersey for away games with the colors reversed, then add a one-line team motto in lettering that matches your mascot's personality.

🎭 Challenge 2: Trick-Shot Show Time

🎯 Performance Art

How can a basketball routine feel as exciting as a circus act or a movie scene? ❓

You're a Globetrotters-style performer. The Harlem Globetrotters are famous worldwide for mixing basketball skill with comedy and showmanship. Now it's your turn to put on a show.

1. Grab a ball (or a rolled-up pair of socks) and a bucket, basket, or box as your hoop.
2. Invent 3 trick moves: a spin, a behind-the-back toss, a dramatic fake-out, or a funny reaction when you miss.
3. Choose a stage name, a costume piece like a headband or cape, and an entrance.
4. Practice the routine 3 times, adding a joke or silly face to every trick.
5. Gather family or stuffed animals as your audience and perform.

🌱 **Make it Easier:** Do just 2 tricks, and let your funny reaction be the main act.

🚀 **Level it Up:** Film a short "movie trailer" for your show first, then perform it live with an announcer voice introducing each trick, and ask your audience to vote on their favorite.

🎵 MUSIC

A basketball is secretly a drum, and a gym full of sneakers is a whole band waiting to play.

🥁 Challenge 1: Dribble Drum Beat

🎯 Rhythm patterns

Can a bouncing ball keep the beat for a whole song? ❓

Turn your ball into the lead drummer! You'll build a four-count rhythm, one layer at a time.

1. Bounce the ball on a hard floor at a steady speed. Count "1, 2, 3, 4" out loud, one bounce per number.
2. Add claps: bounce on 1 and 3, clap on 2 and 4. Repeat until it feels smooth.
3. Add a new layer: tap your knees twice quickly between the counts, like "1 (bounce), and-a (tap tap), 2 (clap)."
4. Try a softer bounce on 3 and a harder bounce on 1. Can you hear the difference?
5. Perform your four-count rhythm live for a family member, repeating it four times without stopping.

Why: A steady beat works like a heartbeat for music. Your ears pick up the pattern, and your body learns to predict when the next bounce is coming.

🌱 **Make it Easier:** Skip the knee taps. Just bounce on counts 1 and 3 and clap on 2 and 4.

🚀 **Level it Up:** Add a second player with a different pattern, then perform both rhythms together so they fit like puzzle pieces.

📣 Challenge 2: Stadium DJ Chant

🎯 Composing chants

What would your team sound like if its cheer was made of squeaks and stomps?❓

You're the stadium DJ, and the crowd is waiting for a hype-up chant. You'll compose one with your own sound effects.

1. Pick a name for your home team and a short chant of 6-8 words, like "Go, go, Hawks, fly high, touch the sky!"
2. Find three body or shoe sounds: rub a sneaker on the floor for a squeak, stomp for a boom, and slap your thighs for a snap.
3. Decide where each sound goes. Maybe a squeak after "go, go," and a double stomp on "fly high."
4. Practice the chant slowly, then speed it up until it feels like game day.
5. Step up to your imaginary DJ booth and perform it for your family, making them stomp and squeak along with you.

🌱 **Make it Easier:** Use a 4-word chant and only two sounds, one stomp and one clap.

🚀 **Level it Up:** Write a second verse that answers the first, like the visiting team's reply, and make the sound effects change between verses.

🫀 BIOLOGY

Your heart and your bones are the hidden engines behind every dribble, jump and slam dunk.

🫀 Challenge 1: Heartbeat Under the Hoop

🎯 Pulse and heart rate

What does your heart do when your body suddenly has to work harder? ❓

You're the sports scientist, and your own heart is today's patient. You'll need a ball, a timer or clock with a second hand, paper and a pencil.

1. Sit still for one minute. Press two fingers on your neck or wrist, count beats for 15 seconds, then multiply by 4. Write it down as "Rest."
2. Dribble as fast as you can for one minute, then stop.
3. Right away, find your pulse and count for 15 seconds. Multiply by 4 and write it as "Right after."
4. Rest for two minutes, then take your pulse again and write it as "Recovered."
5. Draw a simple pulse log with your three numbers. Under it, write one sentence explaining what happened.

Why: working muscles use up oxygen fast, so your heart pumps harder and faster to deliver more blood to them. When you rest, the demand drops and your heartbeat slows down again.

🌱 **Make it Easier:** Count your beats for a full 10 seconds and multiply by 6 instead. Skip the "Recovered" step and just compare resting and right after.

🚀 **Level it Up:** Repeat the drill a second time, but this time try a different drill, like jumping jacks, and see which one raises your pulse more. Write both results in your pulse log.

🦴 Challenge 2: The Tallest Mystery

🎯 Bone growth

What is going on inside a body that keeps growing and growing taller than almost everyone else? ❓

Yao Ming, who played in the NBA, is about 2.29 meters (7 ft 6 in) tall. Let's find out how bones get long. You'll need paper, a pencil and a grown-up who is willing to be interviewed.

1. Ask a grown-up, a librarian, a teacher or a doctor: "How do bones grow longer?" and "What are growth plates?" You can also look in a children's body book or an encyclopedia.
2. Take notes: where are growth plates, what do they do, and what happens when they stop growing?
3. Draw a simple long bone and mark where the growth plates sit near each end.
4. Practice explaining it out loud in your own words.
5. Sit down with a different grown-up and explain how bones grow, using your drawing to help.

🌱 **Make it Easier:** Ask just one question, "What are growth plates?", and explain the answer using only two or three sentences.

🚀 **Level it Up:** Find out what helps bones grow strong (like calcium, vitamin D and movement) and add one tip to your explanation for the grown-up.